Investigating Confidence Intervals for Effect Sizes in One-Way ANOVA: A Comprehensive Analysis of Eta Squared, Omega Squared, and Epsilon Squared

Kasuni Abeykoonge[1*], Johnson Li[1]
Department of Psychology, University of Manitoba

Abstract

Use of effect sizes and their confidence intervals (CIs) has received increasing attention in applied research, yet the performance of CI methods for effect sizes in designs involving a one-way fixed-effect analysis of variance (OF-ANOVA) remains insufficiently understood. This study evaluates CIs, including noncentral F distribution CIs (noncentral F), percentile (Perc) bootstrap, and bias-corrected and accelerated (BCa) bootstrap methods, for eta squared (η^2), omega squared (ω^2), and epsilon squared (ε^2), which are effect sizes commonly used in ANOVA using a Monte Carlo simulation design. We manipulated and evaluated factors across four levels of effect size (0, 0.010, 0.059, 0.138), three levels of sample size (n = 5, 30, 100), and three outcome distributions (normal, positively skewed, uniform) in an OF-ANOVA involving k = 3 groups. The results showed that η^2 exhibited substantial positive bias, particularly in small samples and under the null effect, whereas ω^2 and ε^2 were less biased and showed better coverage as sample size and effect size increased. Perc bootstrap intervals were generally widest, especially in smaller samples, yielding more conservative but less precise interval estimates, while noncentral F and BCa produced comparatively narrower intervals. Coverage for all three CIs deteriorated under skewed distributions and small samples and was further affected by a high proportion of unusable intervals in some conditions. To facilitate transparency and replication, all R simulation code and analysis scripts are openly available via the Open Science Framework (OSF). In line with current recommendations for estimation-based inference, these results provide empirical guidance for selecting, reporting, and interpreting CIs for ANOVA effect sizes under data conditions that researchers commonly encounter in applied work (e.g., small samples and non-normal outcomes).

Keywords: ANOVA, confidence intervals, effect sizes, Monte Carlo simulation

Background

- Estimation-focused reporting encourages researchers to present **effect sizes and confidence intervals**, not only p-values, because they better communicate magnitude, precision, and uncertainty (Cumming, 2014; APA, 2020)
- Prior simulation work has shown that CI performance can change substantially depending on **distribution shape, sample size, estimator, and method of interval construction.**
- In two-group designs, BCa bootstrap intervals performed well under nonnormality, suggesting bootstrap methods may help when assumptions are violated (Kelley, 2005).
- In ANOVA settings (for Root Mean Square Standardized Effect Size (RMSSE)), noncentral-F intervals sometimes outperformed percentile bootstrap intervals, but coverage often deteriorated under skewed or long-tailed distributions (Zhang & Algina, 2008).
- For ω^2 in one-way and factorial ANOVA, coverage of Parametric Perc and BCa. was often below nominal levels, and interval width differed across methods, especially in smaller samples (Finch & French, 2011).

Gap

Existing evidence is fragmented, creating a need for an integrated comparison of CI methods for η^2, ω^2, and ε^2 in OF - ANOVA.

Objective

Systematically evaluate CI performance for η^2, ω^2, and ε^2 in one-way fixed-effect ANOVA across different sample sizes, effect magnitudes, and distribution shapes.

Methods

Design

The simulation studied OF-ANOVA with 3 groups. Conditions varied across:

- 4 true effect size magnitudes (0, 0.010, 0.059, 0.138)
- 5 per-group sample sizes (5, 10, 30, 50,100)
- 3 outcome distributions (normal, uniform, and positively skewed).

Procedures

The study compared noncentral F, percentile bootstrap, and BCa bootstrap CIs for η^2, ω^2, and ε^2. Performance was evaluated using 1,000 Monte Carlo replications per condition and 1,000 bootstrap resamples for bootstrap methods.

Performance Evaluation

Bias, interval usability, coverage, and interval width

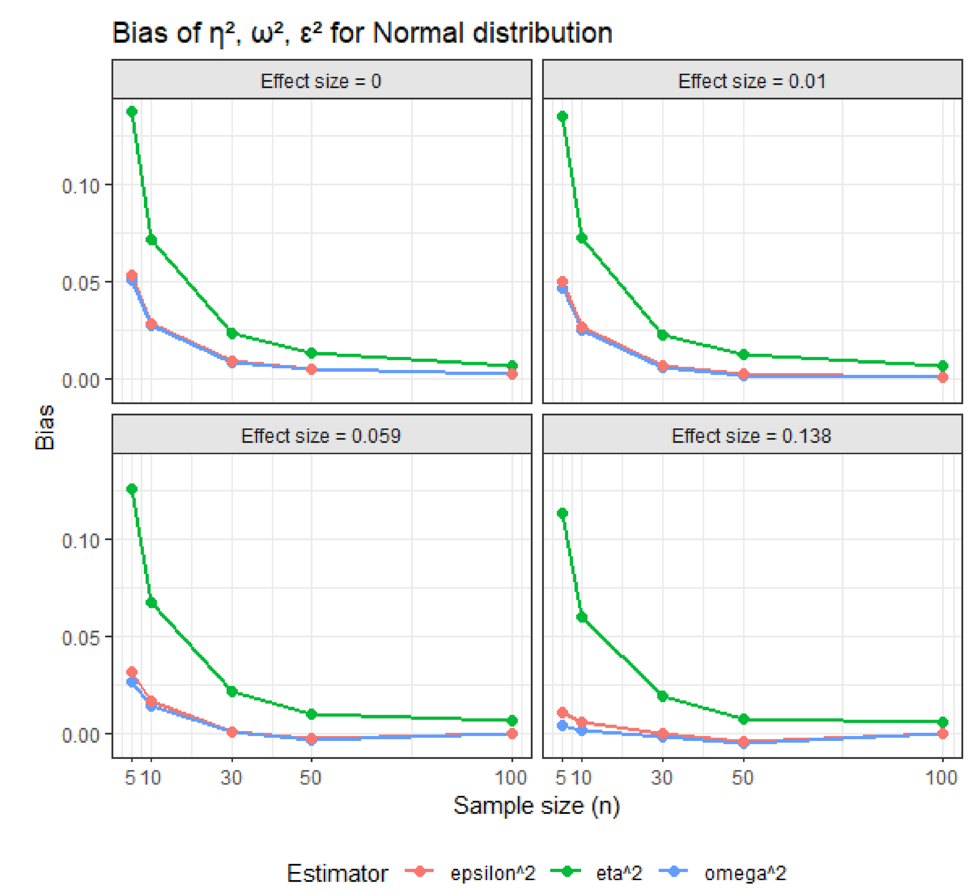
Results

1. Interval usability differed strongly by method

- Percentile bootstrap intervals were essentially always usable across scenarios.
- Noncentral F for η^2 showed occasional failures, mainly in very small samples.
- In BCa failures were negligible
- Under null and small effects, unusable rates for ω^2 and ε^2 were often above 60% for noncentral F and BCa.

2. η^2 showed the greatest positive bias

- Across nearly all conditions, η^2 was more positively biased than ω^2 and ε^2,
- Bias of η^2 improved, and bias of ω^2 and ε^2 approached negligible as the sample size increased.
- The same pattern held across all distributions (normal, uniform, and positively skewed) and effect size magnitudes.



3. Coverage depended on conditions

- Coverage varied by effect size type, effect size magnitude, sample size, and distribution shape.
- Coverage for all three CI methods deteriorated under positively skewed distributions and small samples.
- Conditional summaries sometimes masked poor practical performance when unusable intervals were frequent

4. Width reflected a precision–stability trade-off

- Percentile bootstrap intervals were generally the widest, making them more conservative but less precise. Noncentral F and BCa intervals were typically narrower.
- Interval width decreased steadily as n increased from 5 to 100.
- Widths tended to increased as the true effect grew from null to large.
- Distributional differences were modest relative to sample size and method, though skew often made intervals slightly wider.

Discussion and Conclusion

No single CI method performed best under all conditions. The findings suggest that interval quality should be judged not only by nominal coverage, but also by bias, width, and whether the method can reliably produce usable intervals.

Limitations:

- Specific design conditions
- One-way ANOVA only
- Three CI methods only

References

American Psychological Association. (2020). *Publication manual of the american psychological association* (7th ed). American Psychological Association.

Cumming, G. (2014). The new statistics: Why and how. *Psychological Science*, *25*(1), 7–29. https://doi.org/10.1177/0956797613504966

Finch, W. H., & French, B. F. (2011). A comparison of methods for estimating confidence intervals for omega-squared effect size. *Educational and Psychological Measurement*, *72*(1), 68–77. https://doi.org/10.1177/0013164411406533

Kelley, K. (2005). The effects of nonnormal distributions on confidence intervals around the standardized mean difference: Bootstrap and parametric confidence intervals. *Educational and Psychological Measurement*, *65*(1), 51–69. https://doi.org/10.1177/0013164404264850

Zhang, G., & Algina, J. (2008). Coverage performance of the non-central F-based and percentile bootstrap confidence intervals for root mean square standardized effect size in one-way fixed-effects ANOVA. *Journal of Modern Applied Statistical Methods*, *7*(1), 56–76. https://doi.org/10.56801/10.56801/v7.i.345

SCAN ME

Research Project Webpage: https://maheeshiabeykoon.github.io/Investigating-Confidence-Intervals-for-Effect-Sizes-in-ANOVA/